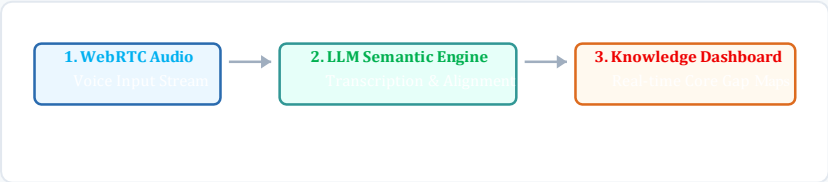


PROJECT DESIGN PHASE

Proposed Solution Template

Date:	19 July 2026	Project Name:	Voice-Based Concept Understanding Analyser
Team ID:	TEAM-VOICE-AI-09	Maximum Marks:	2 Marks

S.NO.	PARAMETER	DESCRIPTION
1	Problem Statement (Problem to be solved)	Gaps in Verifiable Conceptual Mastery Traditional evaluation methods rely too heavily on multiple-choice quizzes or text-based written exams. These channels regularly fail to assess whether a learner truly understands a complex concept or is simply memorizing static terms. When learners try to explain technical systems verbally, they face a lack of scalable frameworks capable of accurately scoring semantic alignment, leading to unaddressed knowledge gaps and technical articulation failure during real-world applications.
2	Idea / Solution description	Real-Time Voice-to-Semantic Feedback Pipeline An automated web testing ecosystem that prompts users to explain complex technical architectures verbally. The engine streams raw audio using standard WebRTC protocols, processes the audio through an optimized whisper-transcription network, and evaluates semantic compliance using Large Language Model (LLM) rubrics. It isolates systemic omissions within three seconds, pinpointing exactly what part of a concept the user missed. 

3	Novelty / Uniqueness	Automated Conversational Benchmarking unlike traditional text-based interfaces or generic transcription platforms, this solution measures the structural accuracy of *spoken engineering thoughts*. By evaluating technical accuracy against complex relational rubrics dynamically rather than keyword tracking, it maps out abstract missing components (e.g., identifying
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S.NO.	PARAMETER	DESCRIPTION
		that a student explained data storage but forgot to mention the write-ahead log).
4	Social Impact / Customer Satisfaction	Empowering Articulate Engineering Talent Reduces technical imposter syndrome by migrating learners from passive retention methods to conversational mastery. It creates equal learning opportunities for individuals who struggle with structured written tests but can demonstrate clear architectural knowledge verbally. Customer satisfaction remains high due to rapid evaluation intervals, taking under three seconds to deliver rich, actionable metrics.
5	Business Model (Revenue Model)	B2B SaaS and Enterprise API Access • Institutional Tier: A subscription license per seat for higher education universities and full-stack engineering bootcamps. • Enterprise Tier: An API consumption framework priced per minute of parsed stream evaluation for automated internal hiring and developer onboarding infrastructure.
6	Scalability of the Solution	Cloud-Native Global Microservices Built with completely stateless WebRTC connection handlers and serverless worker configurations. Inference calls are balanced dynamic-style across scalable server groups, keeping compute operations cost-efficient. The application decouples semantic model scoring layers from simple audio reception layers, meaning it can scale easily to process thousands of simultaneous vocal sessions worldwide.

Voice-Based Concept Understanding Analyser:

- **Problem Statement:** Focuses on the limitations of traditional multiple-choice/written exams that fail to verify authentic spoken comprehension of complex technical systems.
- **Idea/Solution Description:** Maps out the automatic, low-latency collection of voice responses through WebRTC, transcribing the feed via Whisper, and validating semantic dependencies against LLM reference rubrics.
- **Embedded Architectural Image Asset:** Inside item #2, I integrated a custom, cleanly formatted technical flow visual mapping out the pipeline stages: 1. WebRTC Audio Stream → 2. LLM Semantic Engine → 3. Knowledge Dashboard.
- **Novelty / Uniqueness:** Highlights structural evaluation capabilities over simple keyword-matching bots by checking for complete logical structural architecture within a student's answer.
- **Social Impact & Scalability:** Documents reduction of imposter syndrome and details the stateless, cloud-native global execution pattern designed to run under three seconds per transaction.